

CLINICAL PRACTICE UPDATES

AGA Clinical Practice Update on Current Role of Blood Tests for Colorectal Cancer Screening: Commentary



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DESCRIPTION:

Colorectal cancer (CRC) is a leading cause of morbidity and cancer-related mortality in the United States. Despite multiple screening options, adherence to CRC screening in the United States remains suboptimal, and there are racial, ethnic, and geographic disparities in CRC screening rates and outcomes. Advances in diagnostic technology have allowed for development and validation of blood tests for CRC screening. In this Clinical Practice Update, our aims were to review the current evidence on blood tests and the potential implications for CRC screening. We leveraged published modeling studies to understand the optimal test performance characteristics, interval, and uptake that would be needed for blood tests to achieve comparable effectiveness to that of currently available stool tests and screening colonoscopy.

METHODS:

This expert commentary was commissioned and approved by the American Gastroenterological Association (AGA) Institute Clinical Practice Updates Committee and the AGA Governing Board to provide timely guidance on a topic of high clinical importance to the AGA membership. The manuscript underwent internal peer review by the Clinical Practice Updates Committee and external peer review through standard procedures of *Clinical Gastroenterology and Hepatology*.

Background

Colorectal cancer (CRC) remains the third most common cancer in men and women, and the second leading cause of cancer-related death in the United States.^{1,2} Screening is the most effective strategy to reduce the burden of CRC but remains underutilized. Colonoscopy and stool tests are the most commonly used screening modalities in the United States.³ For decades, there has been interest in developing a blood-based CRC screening test. Such tests are now becoming available, thanks to advances in diagnostic technology to detect extremely small amounts of circulating tumor DNA or cell-free DNA shed by a colorectal lesion into peripheral blood.

Novel Blood Tests

Results from population-based clinical validation studies have been reported for 2 blood tests, and 1 blood test is approved by the Food and Drug Administration (FDA) and ready for clinical use. The CRC Shield test by Guardant is a cell-free DNA blood test for CRC screening. The test was validated in a population-based study that enrolled 22,877 individuals across 26 U.S. sites between 45 and 84 years of age who were due for average risk screening.⁴ Participants underwent a blood draw up to

60 days prior to their screening colonoscopy, and for the clinical validation, 7861 individuals were included, of whom 65 had CRC and 1116 had advanced precursor lesions (APLs) (defined as adenoma or sessile serrated lesion ≥ 10 mm, villous histology, high-grade dysplasia, or carcinoma in situ). The sensitivity and specificity of the test for CRC were 83.1% and 89.6%, respectively, with sensitivity for APL of 13.2%.⁴ The test received FDA approval in May 2024, followed by Centers for Medicare and Medicaid Services (CMS) approval, and is commercially available. Table 1 provides a summary of diagnostic performance of the new blood tests compared with currently used stool tests and emerging stool tests (not reviewed in detail here).

The test by Freenome is also based on methylation markers from cell-free DNA in peripheral blood. The clinical validation study across nearly 200 U.S. centers

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Abbreviations used in this paper: APL, advanced precancerous lesion; CMS, Centers for Medicare and Medicaid Services; CRC, colorectal cancer; FDA, Food and Drug Administration; FIT, fecal immunochemical test; MCD, multicancer detection test.

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Table 1. Comparative Diagnostic Performance of Current CRC Tests and Novel Blood and Stool Tests

Modality	Sensitivity			Specificity	Invasive	Endorsed by USPSTF	Covered by CMS	Cost
	CRC	Advanced Precursor Lesion						
Colonoscopy	96%	95%		90%	Yes	Yes	Yes	\$800–\$1500+ (CMS/commercial)
FIT (threshold 20 μ g Hb/g stool)	74%	24%		96%	No	Yes	Yes	\$20–\$25
MT-sDNA (Cologuard)	92%	42% (advanced adenomas and sessile serrated lesions)		87%	No	Yes	Yes	\$500–\$600 (CMS/commercial)
Next-generation MT-sDNA	94%	43% (advanced adenomas and sessile serrated lesions)		93%	No	—	—	Unknown
MT-sRNA (ColoSense)	94%	46% (advanced adenomas)		88%	No	—	—	Unknown
Shield blood test	83.1% (55% for stage 1)	13.2%		89.6%	No	—	Yes	\$1450 list price
Freenome blood test	79.2% (57% for stage 1)	12.5%		91.5%	No	—	—	Unknown

Data are from Lin et al,³⁰ Imperiale et al,³¹ Chung et al,³² Barnell et al,³³ and Shaikat et al.³⁴ CMS, Centers for Medicare and Medicaid Services; CRC, colorectal cancer; FIT, fecal immunochemical test; MT-sDNA, multitarget stool DNA test; MT-sRNA, multitarget stool RNA test; USPSTF, U.S. Preventive Services Task Force.

enrolled 48,995 average risk men and women between 45 and 85 years of age who underwent a blood draw within 120 days prior to a screening colonoscopy. The clinical validation cohort was 27,010 individuals, of whom 72 had CRC and 2567 had APLs. The sensitivity and specificity of the test for CRC were 79.2% and 91.5%, respectively, with sensitivity for APLs of 12.5% (Table 1).⁵ The test is expected to seek FDA and CMS approvals.

Clinical Implications

Although not powered to evaluate differences in sensitivity for CRC by stage, both blood tests reported lower sensitivity for detecting stage 1 CRC compared with stool-based tests, at 55% for Guardant Shield and 57.1% for the Freenome test. The sensitivities for APL were lower than expected, at approximately 13%. These performance characteristics reduce the potential for cancer prevention due to poor APL detection, and for curative treatment due to suboptimal early-stage CRC detection. It is important to note that these are only one-time test performances, and the optimal repeat interval is not known. A main attraction of a blood test is its potential to improve adherence, by bringing individuals into the CRC screening fold who are currently unscreened due to reluctance or inability to undergo stool testing or colonoscopy. At least 1 randomized clinical trial indicates that adherence to blood testing is up to 11% higher than for stool testing.⁶ In clinical practice, the enthusiasm for blood test uptake will likely depend on who the unscreened individuals are for a given practice and their reasons for lack of screening, how the test is offered, convenience vs that of current screening tests, and provider endorsement. An area of need is to understand implementation of blood tests into practice, and the impact on adherence. It is premature to consider interval blood tests between screening colonoscopies.

Multicancer Detection Tests

The ability to detect cell-free or circulating tumor DNA in peripheral blood, and trace it to the organ of origin, has opened up the possibility of early detection of multiple cancers, including CRC, esophageal, gastric, pancreatic, lung, breast, prostate, and others. The field of cancer prevention is aiming beyond CRC detection alone. Multiple companies are pursuing pivotal population-based studies to understand the diagnostic accuracy of multicancer detection tests (MCDs). In an early validation study, the Galleri test by Grail reported an overall sensitivity of cancer detection of 51.5% and specificity of 99.5%.⁷ The company has partnered with the UK National Health Service to recruit 150,000 participants for one of the largest validation studies, and interim analyses are expected in the near future.

Although the concept of detecting multiple cancers with one blood draw appears appealing, many challenges

with MCDs remain. Confirmatory imaging tests can be imperfect, the cost and resource needs for confirming a cancer signal can be substantial, harm is possible, and it is unclear if screening for many cancer types reduces mortality or incidence when factoring lead time and length time bias. While the Galleri test is available in the United States as a laboratory developed test with an out-of-pocket cost, MCDs still require careful evaluation and validation.

Benefit and Cost

It is not realistic to expect evidence for novel CRC screening blood tests from randomized controlled trials with decades-long follow-up to ascertain the critical outcome of CRC mortality. Based on randomized controlled trials of guaiac fecal occult blood testing and sigmoidoscopy,⁸ observational data for fecal immunochemical testing (FIT) and colonoscopy, and an understanding of the adenoma-to-carcinoma sequence, novel CRC screening tests are judged based on their one-time test sensitivity for CRC and APL, and their specificity, compared with an established comparator.⁹ Decision analytic modeling synthesizes data on test performance, CRC epidemiology, and screening's benefits and harms to make long-term projections of benefits and costs with different CRC screening programs.

Rigorous modeling has estimated the long-term impact on CRC incidence and mortality, and total costs, of real-world and hypothetical CRC screening blood tests, with concordance in the high-level results.^{10–13} The most informative real-world cases are the recently approved Guardant Shield test⁴ and the test by Freenome, whose performance characteristics are similar.⁵ The anchor hypothetical case reflects the minimum test performance characteristics set by CMS of CRC sensitivity of 74% and specificity of 90%, which implies a 10% positivity rate for APLs.¹⁴ A critical variable is testing frequency, which is driven by the CMS decision to cover acceptable CRC screening blood tests every 3 years. As can be predicted, test cost is a critical determinant of cost-effectiveness.

Across 4 validated models, annual FIT, a triennial multitarget stool DNA test (Cologuard) and colonoscopy every 10 years were predicted to yield substantially greater reductions in CRC incidence and mortality (eg, 45%–77% and 67%–81%, respectively, for FIT) than triennial Guardant Shield (24%–47% and 53%–59%, respectively) (Table 2).^{10–12} Projections for Freenome and Guardant Shield for reducing CRC incidence and mortality were very similar to each other.¹⁰

At its list price of \$1450, triennial Guardant Shield appears prohibitively costly vs the alternatives, but it appears cost-effective (\$74,800–\$102,300 per quality-adjusted life-year gained)¹⁰ vs no screening. At a test cost of \$300, triennial Guardant Shield could be cost-saving vs no screening, but it would still be less effective and more costly than FIT or colonoscopy.¹⁰

Table 2. Summary Results on CRC Incidence and Mortality Across 4 Validated Decision Analytic Models

	Reduction in Risk of Developing CRC vs No Screening	Reduction in Risk of Dying From CRC vs No Screening
FIT every year	45%–77%	67%–81%
MT-sDNA (Cologuard) every 3 y	40%–73%	64%–78%
Colonoscopy every 10 y	57%–81%	72%–88%
Guardant Shield every 3 y	24%–47%	53%–59%

Data from Ladabaum et al¹¹ and van den Puttelaar et al.¹² CRC, colorectal cancer; FIT, fecal immunochemical test; MT-sDNA, multitarget stool DNA test.

Increasing the frequency of Guardant Shield to biennial or even annual substantially improves its estimated programmatic benefit, but the incremental costs increase much more rapidly than the incremental benefits, making the shorter intervals progressively less cost-effective.¹⁰ These shorter intervals may or may not be covered by payers.

The hypothetical anchor CMS case was very similar to the case of Guardant Shield across models.^{11,12} Hypothetical scenarios that vary test attributes away from the CMS anchor case are highly informative: in order for triennial blood-based CRC screening to approximate the benefits and costs of annual FIT, its APL sensitivity would have to improve substantially (eg, 50%–80%), and its cost would have to be <\$125–\$140.^{11,13}

In the real world, the estimated benefits and costs of a given screening program will be affected substantially by uptake (screening at least once vs never), participation per round through time (consistent vs intermittent), and completion of follow-up colonoscopy after abnormal noninvasive screen.^{10,11,15} It remains to be determined whether improved participation with CRC screening blood tests can compensate for lower expected efficacy and high current costs vs established alternatives. Given that neither FDA approval, CMS coverage determination, nor U.S. Preventive Services Task Force recommendations consider, in aggregate, estimated programmatic effectiveness, benefit/cost trade-offs, and participation,¹⁶ the implementation of novel CRC screening blood tests in real-world practice will rest largely on decisions by health systems, patients, doctors, and payers.

Utilization of Blood Tests for CRC Screening

A screening test's effectiveness in reducing CRC incidence and mortality depends on its clinical

performance (efficacy) and uptake (utilization). The ease of a blood test for CRC screening has raised the possibility of increased utilization. Currently, only 60%–65% of eligible U.S. individuals are up to date with CRC screening, leaving a large gap of unscreened individuals.¹⁷ Reasons for this gap include social determinants of health such as access to care, education, and insurance, but in many cases, patients decline to have a stool test or colonoscopy when offered. In a study of patients who declined other screening, offering a blood test increased CRC screening by 17.5% compared with usual care.¹⁸ There are likely to be 2 groups of participants in a blood test program: persons who decline other forms of screening and persons who will accept a stool test or colonoscopy but substitute the blood test because it seems more appealing.

Modeling studies suggest that if there is substitution of a current blood test for currently recommended stool tests or colonoscopy, then CRC outcomes would be inferior (Table 2).^{11,12} For a blood test to be as effective as FIT or colonoscopy screening, modeling suggests that participation would need to be at least 35% higher. Population outcomes would be similar if for every 3 people who substituted a blood test for a current test there were 2 new screenees. Changes to other parameters that could deliver comparable effectiveness (eg, detection of APLs of $\geq 30\%$ –50%, or annual testing) would require technical improvements in blood tests, or changes in coverage decisions.

Framing the Choice for Patients

The informed decision process may become complicated for primary care providers, who play a key role in CRC screening. If the presentation of screening options is agnostic with regard to effectiveness, patients may elect what seems to be a simple blood test, believing that all tests are equal. However, if patients are told that stool tests or colonoscopy are more likely to detect curable CRC or prevent CRC by detecting important cancer precursor lesions, would patients still choose a blood test?

Disparities in CRC Screening

One of the biggest challenges in CRC is to provide equitable care and minimize disparities. Multiple studies document racial and ethnic disparities in screening. However, a recent study has shown that the gap may be closing between Blacks and Whites.¹⁹ Although screening has improved in all races and ethnicities, studies have indicated the lowest screening rates of 50% in Hispanics.²⁰ It remains to be seen whether blood tests can overcome barriers to screening related to geographic location, cost or cultural norms—and whether they will worsen or lessen disparities.²¹

Rural vs Urban Disparities

Screening uptake and colonoscopy follow-up vary based on access, healthcare settings, and resources. Completing a mailed stool test has remained challenging; one-third or more of eligible patients do not complete FIT.²² A blood test can likely mitigate some of the barriers to screening by making the process less burdensome (eg, removing the need for stool sample collection and shipping). Prior studies have shown that FIT can be effectively implemented in rural settings, although challenges remain.^{23,24} One major challenge is timely colonoscopy completion for positive noninvasive tests. Access could impact utilization. In situations in which access to primary care is limited or geographically distant, a lab visit for a blood draw may be more challenging than collecting a stool sample at home and mailing it. If costs are not fully covered by payers, this would be an additional barrier to access.

Screening With Blood Tests Requires Programmatic Approaches

Blood testing, like other noninvasive testing, requires a programmatic approach. Patients with abnormal tests should complete colonoscopy, and those with normal tests should have repeat testing at an appropriate interval. Prior studies have shown almost 50% of patients do not complete timely colonoscopy within 3–6 months of a positive stool-based test in various healthcare settings.²⁵ Delayed colonoscopies lead to higher CRC incidence and mortality.²⁶ The question remains whether individuals who decline screening colonoscopy will complete a colonoscopy if a screening blood test is abnormal.

Organized healthcare systems in the United States, including health maintenance organizations like Kaiser Permanente, federal facilities like the Veterans Health Administration and Federally Qualified Health Centers, and safety net hospitals have successfully implemented programs of noninvasive tests. A study from Kaiser Permanente showed a significant increase in CRC screening over 15 years and a decrease in CRC incidence and mortality with implementation of an organized program based on mailed outreach with FIT.²⁷ Whether a blood test can be applied as effectively in these settings vs settings with opportunistic screening is not clear.

Screening with a blood test for hepatitis C may provide an informative parallel. U.S. Preventive Services Task Force recommendations to screen “baby boomers” for hepatitis C were announced in 2013.²⁸ This required a one-time blood test. Almost a decade after these recommendations were made, screening uptake has remained variable. Organized healthcare systems such as the Veterans Health Administration achieved a substantial 84% screening rate; however, the screening

rate has remained dismally low (<20%) for community and safety net hospitals.²⁹ Even though hepatitis C and CRC are different diseases in terms of prevalence, awareness, patient cohorts, and testing programs, data on hepatitis C screening may provide a glimpse of the potential successes and failures of public health efforts based on blood tests, and their application in varied healthcare settings.

Conclusions

Despite multiple screening options, adherence to CRC screening in the United States remains suboptimal. Advances in diagnostic technology have led to several blood tests for CRC screening. Guardant Shield is the first FDA and CMS-approved CRC screening blood test in the United States and is available commercially. The sensitivity of current blood tests for APLs and stage 1 CRC is low. It remains to be seen if overall population adherence with CRC screening will improve with the availability of blood tests, vs substitution for colonoscopy or stool tests. Modeling studies project lower benefit of blood tests vs the alternatives in reducing CRC incidence and mortality. Critical factors such as uptake and optimization of repeat testing remain unknown. Patients and doctors should understand that although novel blood tests may improve participation in CRC screening, use of the novel tests could result in net harm if they are substituted for more effective screening tests. For patients who decline any other form of CRC screening, blood tests are an acceptable option.

Summary Points:

- Currently, one blood test (Guardant Shield) is approved by FDA and CMS for CRC screening in average-risk individuals every 3 years.
- Blood tests for CRC screening may improve participation in screening programs because of perceived ease of use.
- Patients should understand that if a blood test is positive, colonoscopy would be recommended.
- Based on one-time test performance characteristics, and long-term modeling projections, programmatic screening with current blood tests every 3 years is substantially better than no screening but would result in lower prevention rates for CRC or CRC death than programmatic FIT, Cologuard, or colonoscopy.
- The first available CRC screening blood tests are best viewed as acceptable in persons who decline the established forms of CRC screening.

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Conflicts of interest

The authors disclose the following: Aasma Shaukat has received royalties from Freenome, Guardant Health, Geneoscopy, Universal DX, and Iterative Health. Uri Ladabaum has served as a consultant for Guardant, Freenome, Inc., Kohler Ventures, Neptune Medical, WilliamBlair, Medacorp, and Mosaic; has stock options with Universal Dx and Cylinder; and has stock with Lean. Priyanka Kanth has a patent on serrated colon polyp RNA panel tissue biomarker (no commercial products of royalties). David Lieberman has served as a consultant for Universal DX, Geneoscopy, and ColoWrap.